



AIMS05-02-005

Issue 2

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AIMS
Airbus Material Specification

Glass Fiber Reinforced Epoxy Prepreg

1543 style glass fabric
(Crowfoot 4-End Satin, 296g/m² dry fiber area weight)

180°C Cure Temperature

Material Specification

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1 Scope

This Material Specification defines the requirements of materials consisting of epoxy resin reinforced with 296g/m² glass fibre (1543 style), having a nominal cure temperature of 180°C, for aerospace application. The requirements specified in this Material Specification shall not be used as stress allowables.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

This specification defines the requirements for two types of material application: Structural and Non-Structural. The different possible applications of glass fibre fabric reinforced epoxy prepregs are defined in the technical specification AIMS 05-02-000 Part 3.

Note: All glass fibre prepregs qualified for a structural application can be used for non-structural application without additional tests.

Materials to be delivered according to the relevant ABS.

3 Normative References

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS 05-02-000Part 1	Airbus Material Specification, Glass fibre reinforced thermosetting prepreg Technical Specification, General Requirements
AIMS 05-02-000Part 3	Airbus Material Specification, Glass fibre reinforced thermosetting prepreg Fabric / Epoxy resin, 180°C curing class Technical Specification, Qualification and batch release testing
AIMS 05-02-000 Part 100	Airbus Material Specification, Glass fibre reinforced thermosetting prepreg Technical Specification, Acceptance testing and Storage life prolongation
ABS 5009	EP-Prepreg. Glass fibre fabric preimpregnated with epoxy resin
AITM 1-0001	Airbus Test Method, Fibre reinforced plastics, Determination of mechanical degradation due to chemical paint strippers
AITM 1-0002	Airbus Test Method, Fibre reinforced plastics, Determination of in plane shear properties ($\pm 45^\circ$ tensile tests)
AITM 1-0003	Airbus Test Method, Fibre reinforced plastics, Determination of the glass transition temperatures
AITM 1-0005	Airbus Test Method, Fibre reinforced plastics, Determination of interlaminar fracture toughness energy, Mode I, G1c test
AITM 1-0006	Airbus Test Method, Fibre reinforced plastics, Determination of interlaminar fracture toughness energy, Mode II, G2c
AITM 1-0007	Airbus Test Method, Fibre reinforced plastics, Determination of plain, open hole and filled hole tensile strength
AITM 1-0008	Airbus Test Method, Fibre reinforced plastics, Determination of plain, open hole and filled hole compression strength
AITM 1-0009	Airbus Test Method, Fibre reinforced plastics, Determination of bearing strength
AITM 1-0010	Airbus Test Method, Fibre reinforced plastics, Determination of compression strength after impact
AITM 1-0019	Airbus Test Method, Fibre reinforced plastics, Determination of tensile lap shear strength of composite joints

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AITM 1-0025	Airbus Test Method, Fibre reinforced plastics, Flatwise tensile test of composites sandwich panels
AITM 3-0001	Airbus Test Method, Fibre reinforced plastics, Analysis of thermoset systems by high performance liquid chromatography
AITM 3-0002	Airbus Test Method, Analysis of non-metallic materials (uncured) by Differential scanning calorimetry
AITM 3-0003	Airbus Test Method, Analysis of non-organic compounds by infrared spectroscopy
AITM 3-0004	Airbus Test Method, Determination of Gel time and viscosity
AITM 3-0006	Airbus Test Method, Fibre reinforced plastics, Determination of material degradation due to chemical products
AITM 3-0008	Airbus Test Method, Determination of the extent of cure by Differential scanning calorimetry
ASTM C 393	Standard Test Method for flexural properties for sandwich constructions
ASTM E 831	Standard Test Method for linear thermal expansion of solid materials by thermo mechanical analysis
EN 2155-12	Aerospace Series, Test Method for transparent materials for aircraft glazing, Part 12 determination of linear thermal expansion
EN 2243-3	Aerospace Series, Structural adhesives test method part 3, peeling test metal-honeycomb core
EN 2329	Aerospace Series, Test Methods for the determination of mass per unit area
EN 2330	Aerospace Series, Textile glass fibre preimpregnated, Test methods for the determination of volatile content
EN 2331	Aerospace Series, Textile glass fibre impregnated, Test methods for the determination of the resin and fibre content of woven textile glass and mass of fibre per unit area
EN 2332	Aerospace Series, Textile glass fibre preimpregnated, Tests method for the determination of the resin flow
EN 2377	Aerospace Series, Glass fibre reinforced plastics, determination of apparent interlaminar shear strength
EN 2743	Aerospace Series, Fibre reinforced plastics, Standard procedures for conditioning prior to testing unaged materials
EN 2747	Aerospace Series, Tensile test properties
EN 2823	Aerospace Series, Fibre reinforced plastics, determination of the effect of exposure to humid atmosphere on physical and mechanical characteristics
EN 2850	Aerospace Series, Carbon thermosetting resin unidirectional laminates, compression test parallel to the fibre direction
EN 3615	Aerospace Series, Fibre reinforced plastics, Procedure for the determination of the conditions of exposure to humid atmosphere and the determination of moisture absorption
ISO 1183	Plastics, method for determining the density and relative density of non-cellular plastics
ISO 1889	Reinforcement yarns, determination of the linear density

4 Technical Requirements

4.1 General

Specification values are minimum except where stated.

4.1.1 Material Description

See table 1

Table 1: Material Description

No.	Characteristic		Requirements
1	Material type		Glass fibre fabric reinforced epoxy prepreg
2	Formulation	Fibre	Glass fibre, Type E
		Fabric	1543 weave US style fabric, 4 harness satin
		Matrix	Modified epoxy resin
		Finish/Sizing	Epoxy compatible
3	Resin content by weight, uncured		37% nominal
4	Material Code		As defined in table 4

4.1.2 Work life and storage life

Worklife and storage life requirements are defined in table 2.

Specific requirements for storage conditions are listed in the relevant IPS.

Table 2: Worklife and storage conditions

No.	Properties	Shop Condition	Symbol	Unit	Requirements
1	Storage life at -18°C	-	-	months	≥ 12
2	Work Life (Tack life) ¹⁾	30°C / 75% R.H.	-	days	≥ 10
		(25±2)°C / (45±10)% R.H	-	days	≥10
Notes: 1) The prepreg can be exposed to an additional minimum of 5 days at the above mentioned shop condition provided that the lay-up is fully bagged and ready for use.					

4.1.3 Processing Data

The curing cycle shall be agreed between the manufacturer and Airbus at the onset of the material qualification program and shall be used to produce all test specimens for qualification testing. The laminates for test purposes shall be cured in an autoclave. The curing cycle shall comply with the general requirements reported in table 3.

The agreed curing cycle shall be specified in the relevant IPS and shall be used for the manufacture of all batch release test specimens.

Table 3: Processing conditions

No.	Cure Parameter	Unit	Requirements
Cure Type: Autoclave			
1	Heating Rate	°C/min	0,2 - 5,0
2	Cure Temperature	°C	180 +10/-5
3	Cure Time	Min	120 - 240
4	Cure Pressure	MPa	0,3 - 1,1
5	Cooling rate	°C/min	0,2 - 3,5
6	Vacuum (abs)	MPa	0,005 - 0,1
Note: The use of an intermediate temperature and low pressure dwell is permissible if agreed between the Manufacturer and Purchaser			

4.2 Specific

4.2.1 Physical-Chemical Properties

Uncured and cured properties, see table 4

4.2.2 Mechanical properties

Cured laminate properties, see table 5

Table 4: Physical-Chemical Properties for all applications

No.	Property		Test Method	Unit	Requirements ¹⁾
Material code					AIMS 05-02-005A
Uncured					
1	Prepreg areal weight		EN2329	g/m ²	470 ± 35
2	Fibre areal weight ²⁾		EN2331	g/m ²	296 ± 25
3	Resin density		ISO 1183 A	g/cm ³	Value to be reported
4	Fibre density		ISO 1889	g/cm ³	2.60± 0,05
5	Volatile content ²⁾		EN2330	%	≤ 1.5
6	Resin content ²⁾		EN2331	%	37 ± 3
7	Physio-chemical tests	HPLC	AITM 3-0001A		(3)
		DSC	AITM 3-0002		
		IR	AITM 3-0003		
		Gel time	AITM 3-0004 B or C		
		Viscosity (resin film)	AITM 3-0004 A		
8	Resin flow		EN2332	%	Value to be reported
9	Tack		As defined in AIMS 05-02-000 Part 1	-	(2)
Cured					
10	CTE ⁴⁾ (Warp and Weft)		EN2155-12	°C ⁻¹	Value to be reported
11	Water Uptake		EN3615 (70°C – 85% R.H. equiv.)	%	≤ 2.0
12	Extent of cure		AITM 3.0008	%	≥ 90
Notes: The stated results are average values 1) 3 specimens per batch 2) Method to be agreed before qualification between Supplier and Purchaser 3) See AIMS 05-02-000 part 1 § 5.4.3, and recorded in the IPS 4) CTE is Coefficient of Thermal Expansion					

Table 5: Mechanical properties for structural application

No.	Property		Test Method	Unit	Conditioning ¹⁾	Test Temp (°C)	Requirements	
							Minimum Average	
Material code							AIMS 05-02-005A	
1	Glass transition temperature	Tg _{onset}	AITM 1.0003	°C	dry	N/A	140	
		Tg _{loss}			wet	N/A	110	
					dry	N/A	Value to be reported	
					wet	N/A		
2a	Tensile strength warp		EN 2747 ²⁾	MPa	dry	-55	650	
						RT	600	
						70	500	
					wet	70	450	
						90	420	
2b	Tensile modulus warp			GPa	dry	-55	30	
						RT	27,5	
						70	22	
					wet	70	21	
						90	21	
2c	Poisson's ratio		EN 2747 ²⁾		dry	RT	Value to be reported	
					wet	70		
						90		
3a	Tensile strength weft			MPa	dry	-55	80	
						RT	60	
					wet	70	50	
						90	48	
3b	Tensile modulus weft			GPa	dry	-55	14	
						RT	12,5	
					wet	70	8,8	
						90	8,2	
4a	Compression strength warp		EN 2850 B	MPa	dry	-55	650	
						RT	600	
						70	440	
					wet	70	620	
						90	560	
4b	Compression modulus warp			EN 2850 B	GPa	dry	-55	27,5
							RT	25
							70	22
						wet	70	30
							90	29
5a	Compression strength weft		EN 2850 B		MPa	dry	RT	200
						wet	70	130
5b	Compression modulus weft				GPa	dry	RT	10
						wet	70	9
(continued)								

(continued)

Table 5: Mechanical properties for structural application (continued)

No.	Property	Test Method	Unit	Conditioning ¹⁾	Test Temp (°C)	Requirements	
						Minimum Average	
Material code						AIMS 05-02-005A	
6a	In plane shear strength	AITM 1-0002	MPa	dry	-55	95	
					RT	85	
					70	72	
					120	62	
				wet	RT	74	
					70	63	
					90	54	
6b	In plane shear modulus	AITM 1-0002	MPa	dry	-55	4,6	
					RT	3,6	
					70	2,7	
					120	2,5	
				wet	RT	2,7	
					70	2,6	
					90	2,5	
7	ILSS warp	EN 2377	MPa	dry	-55	105	
					RT	78	
					70	62	
					120	50	
				wet	RT	66	
					70	54	
					90	46	
8	Tensile lap shear strength	AITM 1-0019 S1	MPa	dry	-55	Value to be reported	
					RT		
					70		
				wet	RT		
					70		
					90		
9	Compression after impact	AITM 1-0010	MPa	dry	RT	Value to be reported	
				wet	70		
10	Fluid Tightness	AITM 2-0037				Value to be reported	
Resistance to aggressive fluids ^{3) 4)}							
11a	Reference	EN2377 + AIMS 05-02-000 part 1: § 4.4.6	MPa	dry	RT	Value to be reported	
				wet	70	Value to be reported	
70°C / 1000h	70			(4)			
11b	Hydraulic fluid				50°C / 1000h	70	(4)
11c	Hydraulic Fluid / Water				RT / 1000h	70	(4)
11d	Fuel				RT / 1h	RT	(5)
11e	MEK						
12	Paint stripper	AITM 1-0001		dry	RT	(5)	
(continued)							

(continued)

Table 5: Mechanical properties for structural application (concluded)						
No.	Property	Test Method	Unit	Conditioning ¹⁾	Test Temp (°C)	Requirements
						Minimum Average
Material Code						AIMS 05-02-005A
Laminate properties (quasi-isotropic)						
13a	Plain tensile strength	AITM 1-0007A	MPa	dry	RT	150
13b	Open hole tensile strength	AITM 1-0007B	MPa	wet	RT	155
					70	95
14a	Plain compression strength	AITM 1-0008A	MPa	dry	RT	290
14b	Open hole compression strength	AITM 1-0008B	MPa	wet	70	240
15	Filled hole tensile strength	AITM 1-0007C	MPa	dry	RT	160
				wet	70	120
16	Filled hole compression strength	AITM 1-0008C	MPa	dry	RT	280
				wet	70	280
17	Bearing strength ultimate ⁶⁾	AITM 1-0009	MPa	dry	RT	660
				wet	70	400
					90	380
Notes: Requirements for properties 2 to 5 and 13 to 16 are calculated with the nominal cured ply thickness of 0,250mm related to a fibre volume content of 46%. Other requirements are calculated with real thickness. 1) Conditioning: 'dry' according to EN 2743; wet is 70°C/85% RH according to EN2823. 2) Type 2 specimens (untabbed) may be used by agreement with Purchaser. 3) Tensile Lap shear strength will have to be tested only in the case of unacceptable failure mode with ILSS test 4) Values after conditioning will have no decrease compared to the reference value (70°C /85% R.H.). tested at the same temperature. 5) Values after conditioning will have no decrease compared to the reference value RT/dry. 6) Values reported at 2%, 6% and ultimate strength						

5 Designation

See relevant ABS.

6 Technical Specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS05-02-000.

7 Incoming Inspection

Prior to use, the materials qualified against this AIMS shall undergo "Technical Incoming Inspection" in accordance with AIMS05-02-000 Part 100'.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 06/06		New standard
2 12/13	Chapter 3 Chapter 7	Part 100 TS added Chapter 7 defined to link to Part 100 specification for Incoming Inspection